SURVEYING

Handwritten Study Notes

Subject : SURVEYING
Code : 23A01301T
Unit : II - Levelling, Contouring, Areas, Volumes

I. Levelling - Introduction

Levelling determines the relative elevations of points. It is done by measuring vertical distances with respect to a horizontal line of sight.

Terms used in levelling

Term	Meaning
Level surface	Surface everywhere normal to the plumb line (e.g. still water surface); curved
Horizontal plane	Plane tangential to the level surface at a point
Datum	Arbitrary reference level; Mean Sea Level (MSL) is the standard datum
Reduced Level (R.L.)	Elevation of a point above or below the datum
Benchmark (B.M.)	Fixed point of known R.L. (GTS, permanent, arbitrary, temporary)
Line of collimation	Line passing through the cross hairs and optical centre of the objective
Axis of telescope	Line joining optical centre and the centre of the eyepiece
Height of Instrument (H.I.)	Elevation of line of collimation above datum
Back sight (B.S.)	First staff reading after setting the level; on a point of known R.L.
Fore sight (F.S.)	Last reading before shifting the instrument
Intermediate sight (I.S.)	Any reading between B.S. and F.S. on the same setting

Change point (C.P.)	Point where both F.S. and B.S. are taken on shifting the instrument
Station	Point where the staff is held

2. Types of Levels

- **Dumpy level:** telescope rigidly fixed to the supports; permanent adjustment; simple and stable; most commonly used.
- **Wye (Y) level:** telescope rests in two Y-shaped supports and can be rotated about its axis, removed and reversed end to end.
- **Tilting level:** telescope can be tilted a little about a horizontal axis by a tilting screw; bubble is brought central for each reading.
- **Automatic (self-aligning) level:** a compensator keeps the line of sight horizontal after rough levelling by a circular bubble; fast and accurate.
- **Digital level:** reads bar-coded staff electronically; data stored automatically.
- **Cooke's reversible level, Cushing's level, Hand level, Abney level, Laser level** – other types for special uses.

Dumpy level – main parts

Telescope (objective, eyepiece, diaphragm with cross hairs), level tube (bubble tube), tribrach with three foot screws, tripod, and a clamp and tangent screw.

Levelling staff

- Wooden or aluminium rod, 3–5 m long, graduated in metres, 0.1 m and 0.005 m (5 mm) divisions. Types: solid, folding, telescopic.
- Self-reading staff (e.g. Sop-with, Philadelphia, Enhanced) and target staff are used.

3. Temporary Adjustments of Level

- **Setting up:** fix the tripod at convenient height with the head approximately level; fix the level on it.
- **Levelling up:** with foot screws, bring the bubble to the centre using the rule of two screws at a time and then the third, in two perpendicular positions.
- **Elimination of parallax:** (a) focus eyepiece until the cross hairs are distinct and black; (b) focus the objective to bring the staff image on the plane of cross hairs.

PARALLAX:
Parallax is present when the image of the staff and cross hairs do not lie in the same plane: the reading changes when the eye moves up or down.

4. Methods of Levelling and Determination of Levels

(A) Height of Instrument (H.P.) Method

H.I. = R.L. of B.M. + B.S.
R.L. of point = H.I. – (I.S. or F.S.)

- Suited to a large number of intermediate points (contouring, cross sections); fewer calculations.
- Arithmetic check: ***Sum of B.S. – Sum of F.S. = Last R.L. – First R.L.***

(B) Rise and Fall Method

Rise or fall = (previous reading) – (next reading)
If positive -> rise; negative -> fall
R.L. = previous R.L. + rise (or – fall)

- Used in differential levelling and longitudinal sections; checks all points, including I.S.
- Check: ***Sum B.S. – Sum F.S. = Sum Rises – Sum Falls = Last R.L. – First R.L.***

COMPARE:
Rise & Fall method has complete check (including intermediate sights). H.I. method has no check on the I.S. readings and is quicker with many I.S.

Sample level book

Stn	B.S.	I.S.	F.S.	H.I.	R.L.
BM	1.525			101.525	100.000
A		1.810			99.715
B	0.890		2.040	100.375	99.485
C			1.250		99.125

Check: sum B.S. = 2.415; sum F.S. = 3.290; difference = -0.875; last R.L. – first R.L. = 99.125 – 100.000 = -0.875. OK.

Types of levelling

- **Simple levelling:** one setting, B.S. and F.S. on two points.
- **Differential levelling:** difference of level between two far-apart points; several set-ups and change points.
- **Fly levelling:** differential levelling to connect a B.M. to the work, only B.S. and F.S.
- **Check levelling:** done to check the earlier work, ends on a starting B.M.
- **Profile (longitudinal) levelling and cross sectioning:** levels along the centre line and at right angles to it for roads, canals.
- **Reciprocal levelling:** used for crossing a river or a valley; eliminates collimation, curvature and refraction errors.
- **Precise levelling:** high-precision, with invar staff, for geodetic control.

5. Reciprocal Levelling

Two instrument positions, one on each bank (near A, then near B). Readings are taken on both staves from both positions.

True difference in level: $h = [(a1 - b1) + (a2 - b2)] / 2$

Combined error $e = [(a1 - b1) - (a2 - b2)] / 2$

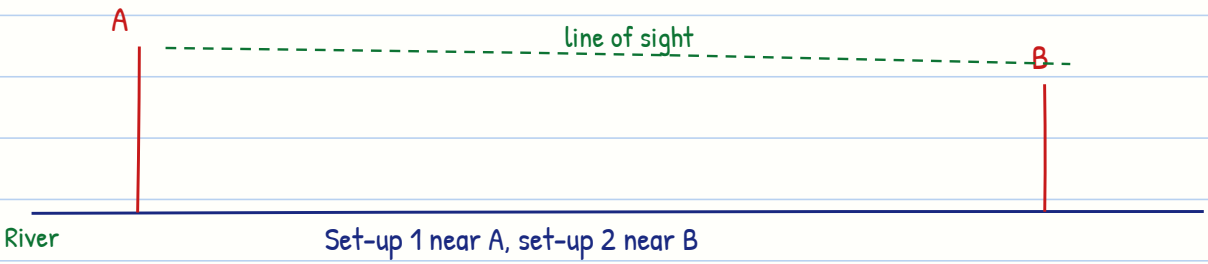


Fig 1: reciprocal levelling

6. Effect of Curvature and Refraction

A level line is curved (follows the earth); the line of sight is straight. So readings at a distance are too high. Also the atmosphere bends light rays downward, which reduces the error a little.

Curvature correction

$$Cc = d^2 / 2R \quad \text{(always subtractive)}$$
$$\text{With d in km: } Cc = 0.0785\, d^2 \text{ metres}$$

Refraction correction

$$Cr = d^2 / 14R = 0.0112\, d^2 \text{ metres} \quad \text{(additive)}$$
$$(Cr = 1/7 \text{ of } Cc)$$

Combined correction

$$C = Cc - Cr = 0.0673\, d^2 \text{ metres} \quad \text{(subtractive)}$$
$$d \text{ in km; } R = 6370 \text{ km}$$

REMEMBER:
Sight at distance d (km): staff reading appears too high by $0.0673\, d^2$ m. So subtract this from the staff reading. For d = 1 km -> 6.73 cm.

Distance of visible horizon (Dip)

$$\text{Dip: } h = 0.0673\, D^2 \quad (h \text{ in m, } D \text{ in km})$$
$$D = \text{sqrt}(h / 0.0673) = 3.855\, \text{sqrt}(h) \text{ km}$$

Example: an observer's eye is at 20 m above sea. Visible horizon: D = 3.855 x sqrt(20) = 17.2 km.

Sources of error in levelling

- **Instrumental:** line of collimation not parallel to the axis of bubble tube; staff graduation error; imperfect parallax adjustment.
- **Personal:** mistakes in reading, booking, holding the staff not vertical, careless levelling.
- **Natural:** curvature, refraction, wind, temperature, settlement of tripod.
- Staff not vertical gives readings too large; ***rocking the staff*** and taking the minimum reading removes this.
- Remedies: keep B.S. and F.S. distances equal, check adjustments (two-peg test), avoid long sights.

Two peg test (collimation error)

Two pegs A and B about 60–100 m apart. Set up level midway (reading a1, b1: correct difference). Then set up near A, take readings a2 and b2. If (a2 – b2) = (a1 – b1) the line of collimation is horizontal; otherwise adjust the diaphragm until the reading b2′ = a2 – (a1 – b1).

7. Contouring

A ***contour*** is an imaginary line joining points of equal elevation on the ground. A ***contour map*** shows the shape of the ground by such lines.

- ***Contour interval:*** vertical distance between two consecutive contours. Depends on scale, relief, purpose and the time and cost permitted.
- ***Horizontal equivalent:*** horizontal distance between two consecutive contours; varies with the slope.
- Typical intervals: 0.5 to 1 m for small scale area, 2–5 m for rugged ground, 10–20 m for a big map.

Characteristics of contours

- Two contours of different elevations ***never intersect*** (except at overhanging cliffs or vertical cliff where they merge or cross).
- Contours of the same elevation never cross; every contour closes on itself within or outside the map.
- Closely spaced contours indicate ***steep slope*** ; widely spaced contours indicate a ***gentle slope*** ; equal spacing means ***uniform slope*** ; parallel, straight, equally spaced lines indicate a plane surface.
- A contour crosses ridge or valley lines at ***right angles*** .
- ***Ring contours*** with higher values inside denote a hill; with lower values inside denote a depression.
- ***U-shaped (convex)*** contours pointing to lower ground represent ***ridges*** ; ***V-shaped*** pointing to higher ground represent a ***valley*** .
- A contour cannot end abruptly; it cannot split into two.

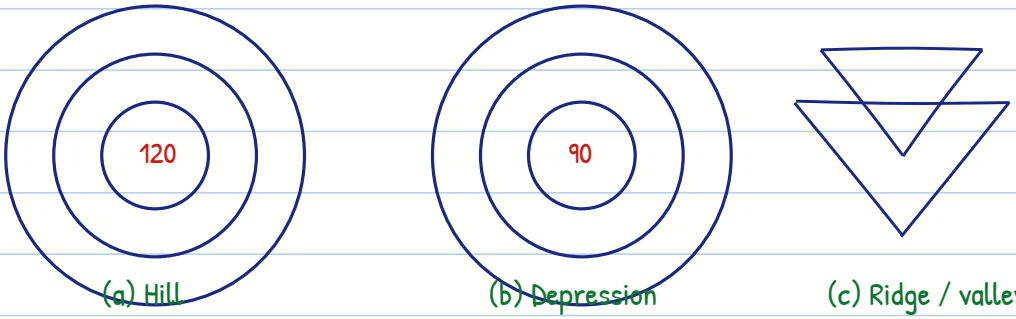


Fig 2: hill, depression and ridge contours

Uses of contour maps

- To study the nature of the ground and choose suitable sites for roads, canals, dams, railways.
- To determine the ***intervisibility*** of two points and draw a section.
- To compute the ***volume of earthwork*** and reservoir capacities, catchment area.
- To trace out ***gradient lines*** (e.g. a road at a given gradient).
- To find the ***drainage and watershed*** areas.

8. Methods of Contour Surveying

Direct method

The contour points are directly located on the ground by levelling, and then surveyed and plotted. Accurate but slow and costly; suited to small areas where accuracy is needed.

- A level is set; B.S. on the B.M. gives the H.I. Required staff reading for the contour of elevation R is: staff reading = H.I. – R.
- The staff man moves up/down the slope until the required reading is obtained; point marked by a peg and then fixed by chain/plane table.

Indirect method

Spot levels of selected points are taken, and contour positions are interpolated. Quicker and cheaper. Mainly for hilly and large areas.

Indirect method	How it works
Squares (grid) method	Area divided into squares (5–20 m); levels at corners; contours interpolated

Cross-section method	Series of lines at right angles to a central line; levels along each
Radial line method	Radial lines from a central point; levels at known distances; best for hills / small areas
Tacheometric method	Stadia hairs give elevations and distances; for hilly regions

Interpolation of contours

- **Estimation:** by eye; rough and only for small works.
- **Arithmetic calculation:** proportions of the level difference and the horizontal distance.
- **Graphical method:** tracing cloth with equally spaced parallel lines; fix the points at the given levels; the lines cut at the intersections give contour points.

Example of arithmetic interpolation: points A (RL 101.4) and B (RL 103.2) are 36 m apart. Contour 102 m is at (102 − 101.4)/(103.2 − 101.4) × 36 = 12 m from A.

9. Areas – Regular and Irregular Boundaries

Areas from field notes (cross-staff survey)

- **Triangle:** A = sqrt[s(s-a)(s-b)(s-c)] where s = (a+b+c)/2; or A = 1/2 b x h.
- **Rectangle:** L x B. **Trapezoid:** 1/2 (a + b) x h. **Circle:** pi r^2.

Areas with an irregular boundary – offsets

Trapezoidal rule:

$$A = d\left[\frac{O_0 + O_n}{2} + O_1 + O_2 + \ldots + O_{n-1}\right]$$

d = equal interval between offsets; O0, O1, … On = the offsets (ordinates). Assumes the boundary between offsets is a straight line.

Simpson's one-third rule (n even, i.e. odd number of ordinates):

$$A = \frac{d}{3}\left[(O_0 + O_n) + 4(O_1 + O_3 + \ldots) + 2(O_2 + O_4 + \ldots)\right]$$

CONDITION:
Simpson's rule: number of ordinates must be ODD (even number of strips). Boundary assumed to be a parabola; more accurate than trapezoidal. If not odd, apply Simpson to the first strips and the trapezoidal rule to the last one.

- **Average ordinate rule:** A = (sum of ordinates / number of ordinates) x length of base.
- **Mid-ordinate rule:** A = d x (sum of mid-ordinates).

Area from coordinates (closed traverse)

2A = Sum [Xi (Y(i+1) - Y(i-1))] = Sum of products of X and Y cross

- **Planimeter:** mechanical instrument: tracing the boundary gives area; Area = M (final - initial reading + 10 N when zero is crossed). **Digital planimeter** gives area directly.
- Area by dividing into **triangles** from a map, or by **graph (squares)** method counting squares.

Example (trapezoidal and Simpson)

Offsets at 10 m interval: 3.0, 4.2, 5.1, 4.8, 3.6 (m). Trapezoidal: A = 10[(3.0+3.6)/2 + 4.2 + 5.1 + 4.8] = 10[3.3 + 14.1] = 174 sq.m.

Simpson: A = 10/3 [(3.0+3.6) + 4(4.2+4.8) + 2(5.1)] = 10/3 [6.6 + 36 + 10.2] = 176 sq.m.

10. Volumes - Earthwork

Volume of earthwork in **cutting** (excavation) and **embankment** (filling) is needed for estimating the cost of roads, railways and canals.

Level section (formation level horizontal, ground level across the section)

Area = (b + s h) h (b = formation width, s : 1 side slope, h = depth/height at centre)

Cross-section for a level section: A = b h + s h^2

Two-level and three-level sections

When the ground slopes across the section, the cross-section is split into triangles and trapezoids (using the side slopes to find where the slopes meet the ground), and the

areas are added. The side-slope toe distance on each side = s x (depth at that side).

Calculation of volume

Method	Formula
Trapezoidal (average end area)	V = d [(A1 + An)/2 + A2 + ... + A(n-1)]
Prismoidal formula	V = d/3 [(A1 + An) + 4 (A2 + A4...) + 2 (A3 + A5...)]
Prismoidal correction (level section)	Cp = (s d / 6)(h1 - h2)^2
Mean area method	V = (A1 + A2)/2 x d (small segments)

- ***Trapezoidal rule*** is simpler but gives ***larger*** (overestimate) volume; the prismoidal formula is more exact and needs an odd number of sections.
- Prismoidal correction is ***subtracted*** from the trapezoidal volume: V(prism) = V(trap) - Cp.

Example

Level section, formation width 10 m, side slope 2:1. Length 20 m; depth at A1 = 1.5 m, A2 = 2.5 m. Area A1 = (10 + 2 x 1.5) x 1.5 = 19.5; A2 = (10 + 2 x 2.5) x 2.5 = 37.5. Volume = (19.5 + 37.5)/2 x 20 = 570 cu.m.

II. Capacity of Reservoirs

The storage capacity of a reservoir at a site is obtained from a ***contour map*** of the area. Area enclosed by each contour (found with a planimeter) is computed, and the volume between contours is calculated.

Trapezoidal: V = h [(A1 + An)/2 + A2 + A3 + ... + A(n-1)]

Prismoidal: V = h/3 [A1 + An + 4 (A2 + A4 + ...) + 2 (A3 + A5 + ...)]

- h = contour interval; A1, A2... areas enclosed by successive contours from the bottom to FRL (full reservoir level).
- ***Cone formula*** between two contour areas: V = h/3 (A1 + A2 + sqrt(A1 A2)) - more accurate for gradually changing sections.
- Plot an ***elevation - capacity curve*** and an ***elevation - area curve*** to know the storage at any water level.

- Dead storage, useful storage and flood storage are the three zones considered.

12. Benchmarks and Field Procedure

- **G.T.S. benchmark:** Great Trigonometrical Survey benchmark, fixed with great accuracy, a permanent stone pillar with R.L. above MSL.
- **Permanent B.M.:** fixed by govt. departments (PWD, railways) on stable structures such as culverts, bridge piers, building plinths.
- **Arbitrary B.M.:** assumed R.L. (e.g. 100.000 m) used for small works where MSL is not known.
- **Temporary B.M.:** established at the end of a day's work, to continue the next day.

Field procedure of differential levelling

- Set up the level midway between B.M. and the first change point; level up; remove parallax.
- Take B.S. on the B.M.; then F.S. on the change point (the staff holder keeps the staff on the same point while the level is shifted).
- Move the level ahead, take B.S. on the same C.P., and continue to the last point.
- Keep B.S. and F.S. distances approximately equal; maximum sight length about 100 m.
- Close the work on a known B.M. (check levelling) and verify the closing error: allowable error = 12 sqrt(K) mm for ordinary work, K in km.

13. Worked Problems

Problem 1: Rise and Fall method

Staff readings on consecutive points: 2.285 (B.S. on B.M. of RL 200.000), 1.650, 2.100, 1.325 (F.S.), 1.860 (B.S.), 0.950 (F.S.). Find the RL of all points.

Pt	B.S.	I.S.	F.S.	Rise	Fall	R.L.
BM	2.285					200.000
1		1.650		0.635		200.635
2		2.100			0.450	200.185
3 (C.P.)	1.860		1.325	0.775		200.960
4			0.950	0.910		201.870

Check: sum B.S. = 4.145; sum F.S. = 2.275; difference = +1.870; sum rise − sum fall =

2.320 − 0.450 = +1.870; last RL − first RL = 201.870 − 200.000 = +1.870. All three agree.

Problem 2: Curvature and refraction

Find the combined correction and the true difference in level if the staff reading at 800 m distance is 2.450 m and the instrument height is 1.520 m.

$$C = 0.0673 \times (0.8)^2 = 0.0431 \text{ m}$$

$$\text{Corrected reading} = 2.450 - 0.043 = 2.407 \text{ m}$$

$$\text{True fall} = 2.407 - 1.520 = 0.887 \text{ m}$$

Problem 3: Gradient line from a contour map

To fix a road at gradient 1 in 25 with contour interval 2 m, the horizontal distance between two successive contours is 25 × 2 = 50 m. Set a divider to 50 m on the map scale and step from contour to contour.

$$\text{Horizontal equivalent} = \text{contour interval} \times (\text{gradient denominator})$$

$$= 2 \times 25 = 50 \text{ m}$$

Problem 4: Prismoidal formula

Three sections 30 m apart have areas 40, 52 and 60 sq.m. Find the volume.

$$\text{Prismoidal: } V = 30/3\,[(40 + 60) + 4 \times 52] = 10 \times 308 = 3080 \text{ cu.m}$$

$$\text{Trapezoidal: } V = 30\,[(40+60)/2 + 52] = 3060 \text{ cu.m}$$

Here the trapezoidal value differs by only 0.65%; in normal earthwork the difference is small, so trapezoidal volumes are usually accepted.

Problem 5: Reservoir capacity

Areas enclosed by contours at interval 5 m: 200, 1400, 3200, 5800, 9000 sq.m (with contour 0 at the bottom). Find the storage up to the top contour using the trapezoidal formula.

$$V = 5\,[\,(200 + 9000)/2 + 1400 + 3200 + 5800\,]$$

$$= 5\,[\,4600 + 10400\,] = 75{,}000 \text{ cu.m}$$

14. Planimeter – How the Area is Read

- A ***planimeter*** has a tracing arm, a pole arm, a roller (measuring wheel) with a graduated disc and a vernier.
- Place the pole outside the figure, trace the boundary clockwise with the tracer point, and note the initial and final reading of the wheel.

$$\text{Area} = M\,(F - I + 10N + C)$$

M = multiplying constant, N = no. of times zero mark passes the index,
C = constant of the instrument (when the pole is inside the area)

- Use: quick area of an irregular boundary or of contour-enclosed areas for reservoir capacity. Accuracy about 1 in 200.
- Verify by tracing a known area (e.g. circle or a square) to calibrate the planimeter.

15. Important Questions to Prepare

- Differentiate between rise and fall method and H.I. method with a neat level book.
- Derive the combined correction due to curvature and refraction.
- State the characteristics of contours; explain the direct and indirect methods of contouring.
- Explain the trapezoidal and Simpson's rules for area; derive the prismoidal formula.
- Compute the capacity of a reservoir from a contour map.

16. Quick Revision

- H.I. = R.L. + B.S.; R.L. = H.I. – F.S.; Check: sum B.S. – sum F.S. = last RL – first RL.
- Combined curvature and refraction: 0.0673 d^2 m (subtractive).
- Contours never cross (except cliff); steep = close; gentle = wide; closed rings = hill / depression.
- Trapezoidal rule, Simpson's 1/3 rule (odd ordinates), prismoidal formula (odd sections).